Kaustubh Hemant Sonawane

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UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

STATEMENT

Backend Engineer with 3 years of industry experience building **Python** APIs and data-heavy backends for regulated financial services: **FastAPI** microservices, **PostgreSQL** schemas with SQL functions and procedures, and **Morningstar** ingestion pipelines, deployed on **AWS** with **Docker** and a one-click **Jenkins** and **GitLab CI/CD** pipeline. Designed the first services as the only engineer alongside my manager, later as de facto team lead for up to 4 interns, and hardened them (**Nginx** reverse proxy, **RBAC** over **LDAP**, **AWS Security Groups**) until they passed the security team's VAPT after repeated failures. Reaches for the faster option when performance matters: served fund-comparison calculations from **ExpressJS** APIs during development because the Python APIs were slower. Completed an MSc in Data Science (awaiting result), with a dissertation project with AstraZeneca built as FastAPI microservices; twice recognised with Best Team Project.

EDUCATION

University of Bristol – MSc. Data Science (awaiting result) September 2025 – September 2026
NMIMS B.Tech Information Technology (Dual Degree) June 2017 – April 2022
NMIMS MBA Technology Management (Dual Degree) June 2017 – April 2022

WORK EXPERIENCE

HDFC Asset Management Company Ltd. July 2022 – July 2025

AI Solutions Engineer / Forward Deployed Engineer

APIs & Backend Services

- Specified, designed and built the backend for the company's first GenAI applications, initially with only my manager: **Python FastAPI** REST API **microservices** (**uvicorn**) on their own **AWS EC2** instance, serving a **ReactJS** frontend on a separate one, for Compliance, Client Service and Investment. Backed them with **RAG** (**LlamaIndex**, **Azure OpenAI** GPT-3.5 & Llama-3) over a separate **pgvector** database per department, so each team retrieved only from its own documents.
- Built **tool-calling** agents whose **Python** functions run **SQL** queries to compare funds and generate PDF reports on request; built **ExpressJS** APIs during development to serve the fund-comparison calculations because the Python APIs were slower, with the final backend in **FastAPI**. Released to real users and changed the services based on what they found.
- Weighed cost against quality when placing models: **self-hosted Llama-3** on a **GPU EC2** instance for the largest department and the **Azure OpenAI** API for the others, with the ability to switch models as needs changed.

Data Pipelines & Databases

- Designed the **Normalized SQL Schema** (**PostgreSQL**) for the internal analytics platform, with **Optimized SQL** queries and **SQL Functions** and **Procedures** for efficient data insertion; wrote **Python** ingestion scripts for the **Morningstar API** covering every fund in the Indian mutual fund market (2023–2025), refreshed fortnightly and monthly and integrated with the legacy analyst data.
- Built a **Python** pipeline, scheduled with **EventBridge**, **Lambda** and **AWS SageMaker**, that uses **Azure OpenAI** to parse over **300** daily emails from credit rating agencies into structured data (first version used **AWS Textract**, replaced when a rating agency began sending emails as images), fuzzy-matches company names against the ones analysts track and sends a daily summary, saving **60** analyst hours a month.

- Migrated financial analysts from Excel to **Power BI** dashboards, improving efficiency by **10** hours a week, then recreated them in the **ReactJS** frontend because Power BI licensing was too costly to scale to the nationwide Client Service team; analysed customer data with **Python Pandas**.

Reliability, Security & Delivery

- Packaged the **FastAPI** microservices (uvicorn) as **Docker** containers on **AWS** (EC2, ECR) and automated deployment of the separate frontend and backend repositories (dev and prod) through a one-click **Jenkins** and **GitLab CI/CD** pipeline.
- Protected the internal applications with **RBAC** using a **Python** authentication function against the company **LDAP** server and **AWS Security Groups** limiting access to company private IPs; deployed an **Nginx** reverse proxy, one of the fixes that took the applications from repeatedly failing to passing the security team's **VAPT**, and coordinated with them on which ports to whitelist.
- Worked directly with Client Service, Compliance and Investment to turn requests into clear requirements and feedback into releases; acted as de facto team lead for up to 4 interns in my final year; recognised with Best Team Project twice (2024 rating-email automation, 2025 department RAG chatbot).

PROJECTS

- **Wingman:** AI dating-coach web app on a Python FastAPI backend (ReactJS frontend) calling LLMs via NVIDIA NIM. Rate limits shaped the design: calls serialised behind a semaphore, 12 evaluations instead of 108, results streamed to the UI over server-sent events and cached by input hash; LLM output treated as untrusted and validated with Pydantic. GitHub: https://github.com/Arishtanemi4/wingman.git
- **University of Bristol MSc. Data Science Dissertation in partnership with AstraZeneca:** CellLineSelector (Team project, co-built. A tool built as FastAPI microservices with a React frontend that ranks cancer cell lines by target inclusion and exclusion genes and returns the evidence behind each pick. Personal focus: the data preprocessing pipeline that joins raw genomic sources into clean tables with integrity checks, the scoring algorithm and the RAG layer; refined with feedback from AstraZeneca scientists.) GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/

ACHIEVEMENTS

- **Encode Vibe Coding Hackathon:** Codeplain.ai track 2nd place winner. (Wingman.ai – An AI powered dating app simulation using codeplain's spec-driven framework) | GitHub: https://github.com/Arishtanemi4/wingman-codeplain.git
- **Project Hack 27**: Rolls Royce Derby 4th Place winner. (Human Centric Data Analytics tool to predict the competition winner based on human wellbeing and psychological temperament survey data.) | GitHub: https://github.com/Arishtanemi4/team-4b.git
- **University of Bristol Datathon sponsored by Lloyd's Bank:** 3rd Place winner. (Kaggle Credit card fraud detection problem.)

SKILLS

- **Tools:** REST APIs, Git, Docker, Jenkins, GitLab (CI/CD), AWS (EC2 including a GPU instance, S3, ECR, Lambda, EventBridge, SageMaker), Nginx, Linux, LDAP, Azure OpenAI, LlamaIndex, Claude Code, Power BI, VS Code.
- **Languages:** Python (FastAPI, uvicorn, Pydantic, pandas, scikit-learn), SQL, JavaScript/TypeScript (NodeJS express, ReactJS), R (dplyr, tidyverse, ggplot2), Shell
- **Databases:** PostgreSQL (pgvector semantic search, SQL functions and procedures), SQLite